CSC 2050 - Week 10 Activity

For this week you will create a part of a file system called i-node. I-nodes are the parts that are closer to the files and the actual data stored in file systems. They are responsible to hold information about the types of the files they represent, and where the data for those files are.

Different file systems may hold different data and at a different places in an i-node. Examples you can find in you book, in the slides on CANVAS, and online *https://man7.org/linux/man-pages/man7/inode.7.html*

For this activity, the i-nodes should keep track of the following information,

- **I-node ID**: A unique ID for the i-node.
- **File Type**: Tells the system if it is a regular file, a directory, a symbolic link, etc.
- **File Size**: The size of the file in bytes.
- **Ownership IDs**: The Numeric User ID (UID) and Group ID (GID) of the owner.
- **Permissions**: Read, write, and execute permissions (e.g., rwxr-xr-x).
- **Timestamps**: Tracking when the file was last accessed (atime), modified (mtime), or when its metadata changed (ctime).
- **Hard Link Count**: Tracks how many filenames (directory entries) point to this exact inode.
- **Block Size**: The size of each data block. We'll use files for our data blocks, so this is going to be the numbers of characters in the file. Use 1024.
- **Block Offset**: The bytes from the beginning of the files where the data are stored.
- **Data Block Pointers**: Addresses pointing to the actual physical locations (blocks) on the storage drive where the raw data is kept. This should be a 15-element array from which the first 12 elements are direct pointers, the next 2 are double in-direct pointer and the last one triple-indirect pointer. Each of these pointers should point to separate files so data can be stored in those files.

The i-nodes should have some basic functionality.

- The **mkDir()** function should create a directory i-node.
- The **mkFile()** function should create a file i-node.
- The **writeFile()** function should write data in an i-node.
- The **readFile()** function should read data from an i-node.

When everything is implemented, write test cases in your main to test the functionality of your i-node data type.

Submission: Submit your .c file on CANVAS on W10 activity. Make sure there are no warnings during the compilation.